Tobit Regression Analysis Report

Automated Analysis Pipeline

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Abstract

This report documents the full Tobit analysis workflow for the Florida experiment dataset, including materials, data cleaning, long-format transformation, bounded-outcome modeling, statistical interpretation, assumptions, and conclusions.

1 Introduction

This report evaluates how empathy and social identity relate to moral judgments of negotiators in an incentivized experiment. The reporting workflow is designed to be reproducible and publication-oriented, using the same fitted models that drive the Markdown and Word outputs.

2 Materials and Methods

2.1 Inputs and Materials

Dataset: data_final_FLORIDA.xlsx Codebook: datacard.md Hypotheses source: hypotheses.md . Materials. The analysis uses the workbook data_final_FLORIDA.xlsx, the project codebook datacard.md, and the substantive framing in hypotheses.md. The source workbook contains 63 participant rows and repeated scenario blocks for 10 stages, with two negotiator decisions and two numerical judgments per stage. Methods. The pipeline reshapes the workbook from wide participant format to long negotiator-level format, scores the Interpersonal Reactivity Index, filters the primary analysis sample using the attention checks, estimates Tobit regressions for the bounded outcome, and exports tables, figures, Markdown, Word, LaTeX, and PDF reports.

2.2 Sample and Data Structure

The workbook contains 63 participants, of whom 58 passed both attention checks. The primary analysis retains 58 participants and 1160 negotiator-level judgments. The Tobit hypothesis models focus on 569 harmful decisions in which a negotiator accepted the payoff-increasing but victim-harming deal.

Table 1: Participant summary.

Metric	Value
Participants in workbook	63.000
Participants passing both attention checks	58.000
Participants in primary analysis	58.000
Mean age (analysis sample)	20.293
SD age (analysis sample)	4.877
Women in analysis sample	27.000
Men in analysis sample	31.000
Humanities in analysis sample	27.000
Engineering in analysis sample	31.000
Victim-first treatment in analysis sample	29.000
Observer-first treatment in analysis sample	29.000

2.3 Data Cleaning and Step-by-Step Data Handling

Data cleaning proceeded in explicit steps. Step 1: validate that all required participant-level and scenario-level columns are present in the workbook. Step 2: preserve the raw participant-level file, generate stage-specific role indicators, and derive IRI composite and subscale scores. Step 3: compute the attention-check flag and define the primary analysis sample as participants who passed both checks, had a non-missing IRI composite, and are in treatment 1 or 2. Step 4: reshape the repeated scenario variables into negotiator-level long format, yielding two judgment rows per stage per participant. Step 5: derive perpetrator alignment, victim alignment, same-group harm, and harmful-decision indicators. In this dataset, 0 participants were excluded for treatment = 0, 5 participants fail at least one attention check, and 0 participants have a missing IRI composite, leaving 58 participants in the primary sample.

2.4 Variable Transformation and Analysis Setup

Variable handling and transformation were kept close to the observed data structure. The dependent variable is the raw judgment score, bounded between -9 and 9, and it is modeled directly rather than transformed into a severity index. The empathy predictor is standardized as iri_total_z so its coefficient reflects a one-standard-deviation change in empathy. The main hypothesis models are restricted to harmful decisions where decision_accept = 1, because the hypotheses concern condemnation of harmful conduct rather than neutral or prosocial choices. Control variables include observer role, participant faculty, sex, age, socioeconomic status, stage indicators, and negotiator-slot indicators.

2.5 Word-Friendly Model Equations

Observed outcome definition

y_ij = -9 if y*_ij <= -9
y_ij = y*_ij if -9 < y*_ij < 9
y_ij = 9 if y*_ij >= 9

Main harmful-decision model

y*_ij = beta0 + beta1*IRI_i + beta2*OutgroupPerp_ij + beta3*ControlPerp_ij + beta4*VictimOutgroup_ij + beta5*(IRI_i*OutgroupPerp_ij) + beta6*(IRI_i*ControlPerp_ij)

+ gamma'*X_ij + e_ij

Same-group-harm model
y*_ij = alpha0 + alpha1*IRI_i + alpha2*SameGroupHarm_ij + alpha3*OutgroupPerp_ij + alpha4*VictimOutgroup_ij + delta'*X_ij + u_ij

2.6 Empathy Scale Quality

Table 2: Empathy scale summary.

Scale	Items	Mean	SD	Min	Max	Alpha
IRI total	28	2.196	0.448	1.036	3.250	0.767
Fantasy	7	2.111	0.723	0.286	4.000	0.644
Empathic concern	8	2.222	0.485	1.125	3.125	0.299
Perspective taking	7	2.488	0.665	0.571	3.714	0.597
Personal distress	6	1.918	0.635	0.000	3.000	0.521

3 Results

3.1 Exploratory Data Analysis

We first explore the raw dependent variable, showing its bounded nature and the piling up at the endpoints which justifies the Tobit approach.

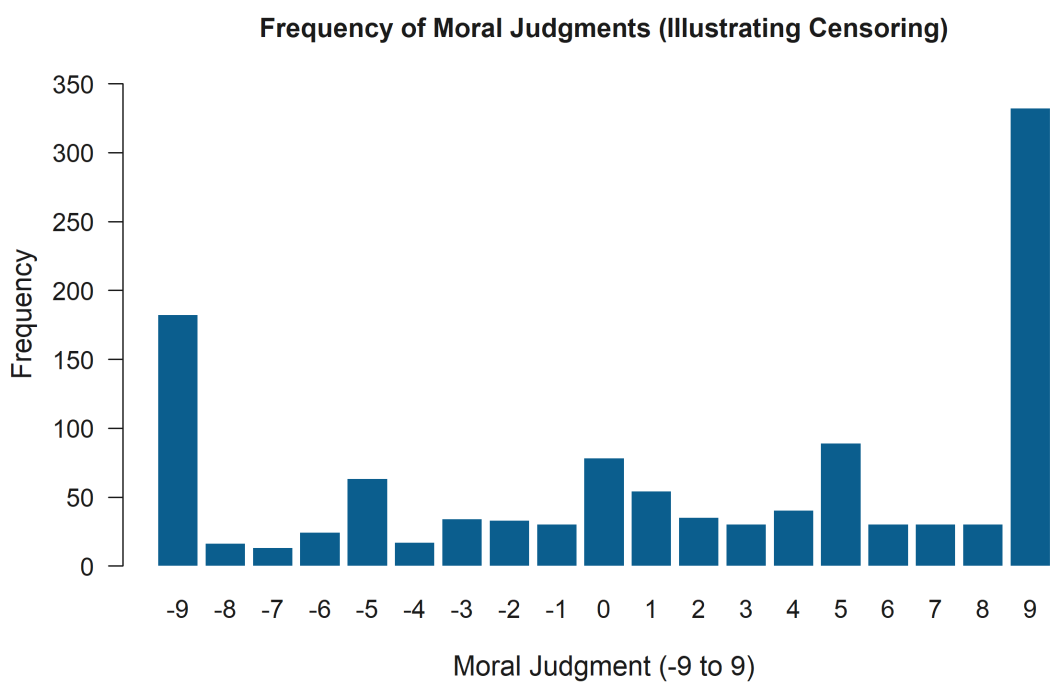

Figure 1: Frequency of moral judgments illustrating censoring.

Additionally, the following table and scatterplots summarize the relationship between moral

judgments and each empathy subscale.

Table 3: Pearson correlations between judgements and empathy subscales.

Subscale	Pearson_r	P_value	CI_low	CI_high
iri_total	0.008	0.785	-0.050	0.066
iri_fs	0.005	0.856	-0.052	0.063
iri_ec	-0.092	0.002	-0.149	-0.034
iri_pt	0.042	0.154	-0.016	0.099
iri_pd	0.061	0.037	0.004	0.118

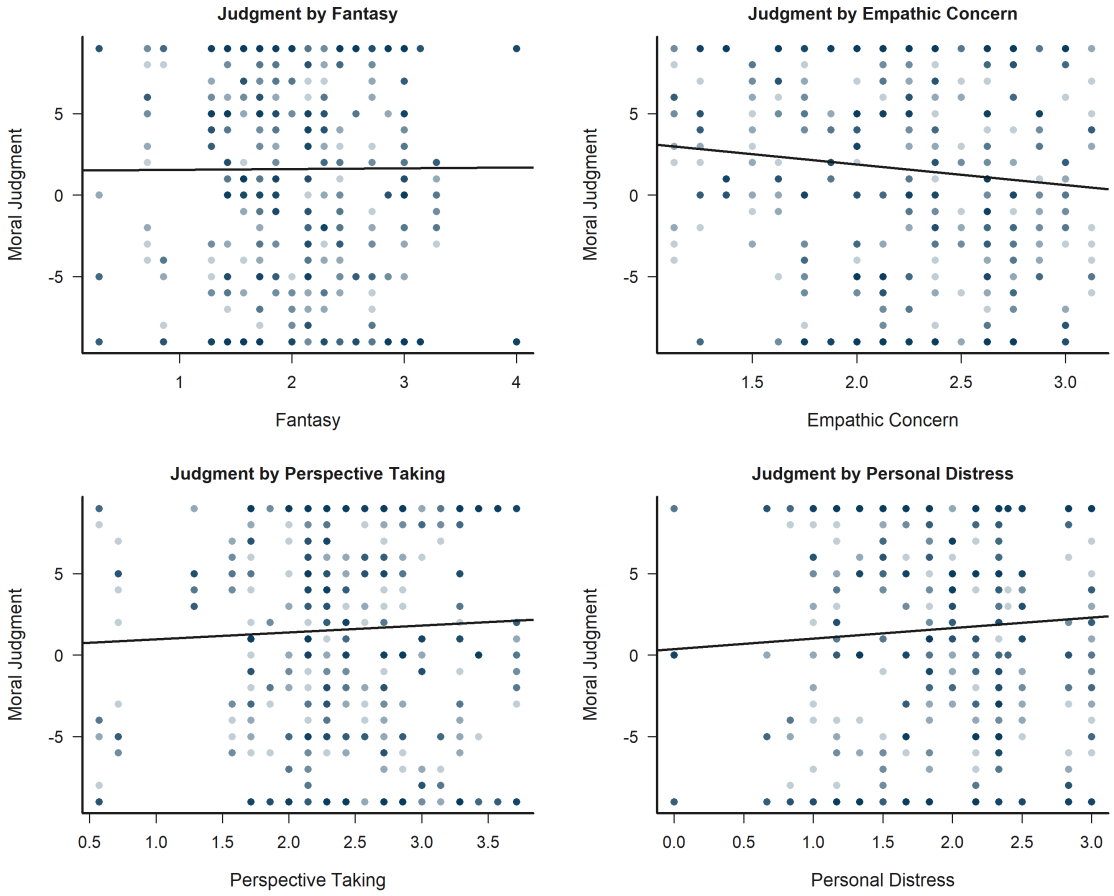

Figure 2: Scatterplots of judgment by IRI subscales.

3.2 Descriptive Results

The observed Tobit outcome is the raw moral-judgment score from -9 to 9, where lower values indicate harsher condemnation. Descriptively, accepted harmful deals receive much lower ratings than rejected deals (mean judgment -3.19 versus 6.21). Within the harmful-decision sample, same-faculty harm averages -3.55 judgment points compared with -2.91 for cross-faculty harm. Outgroup perpetrators average -3.12 judgment points versus -3.34 for labeled ingroup perpetrators.

Table 4: Judgment summary.

Metric	Value
Negotiator-level judgments in analysis sample	1160.000
Harmful decisions in primary model sample	569.000
Acceptance rate in analysis sample	0.491
Mean raw judgment for rejected decisions	6.212
Mean raw judgment for accepted decisions	-3.192
Left-censored share at -9 in accepted sample	0.299
Right-censored share at 9 in accepted sample	0.047

Table 5: Harmful-decision descriptive summary by group and role.

negotiator_alignment	role	Observations	MeanJudgement	SDJudgement	SEJudgement	Lower95
ingroup	victim	95	-3.979	4.842	0.497	-4.953
outgroup	victim	97	-3.289	5.278	0.536	-4.339
control	victim	104	-3.740	5.593	0.548	-4.815
ingroup	observer	92	-2.685	5.056	0.527	-3.718
outgroup	observer	80	-2.913	5.406	0.604	-4.097
control	observer	101	-2.475	5.723	0.569	-3.591

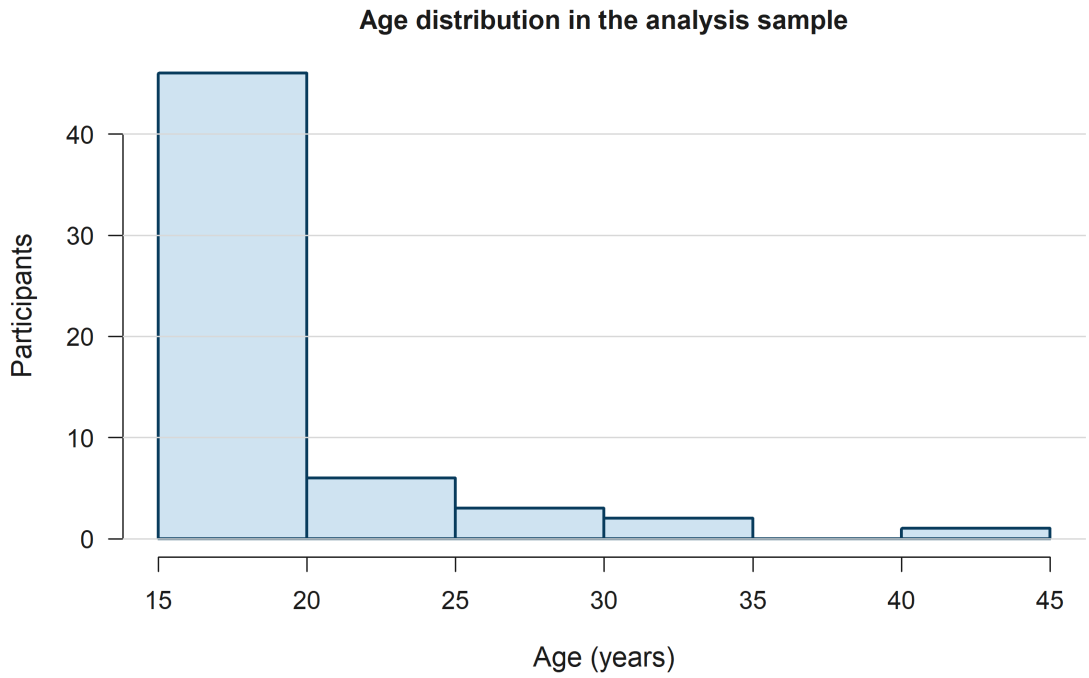

Figure 3: Age distribution in the analysis sample.

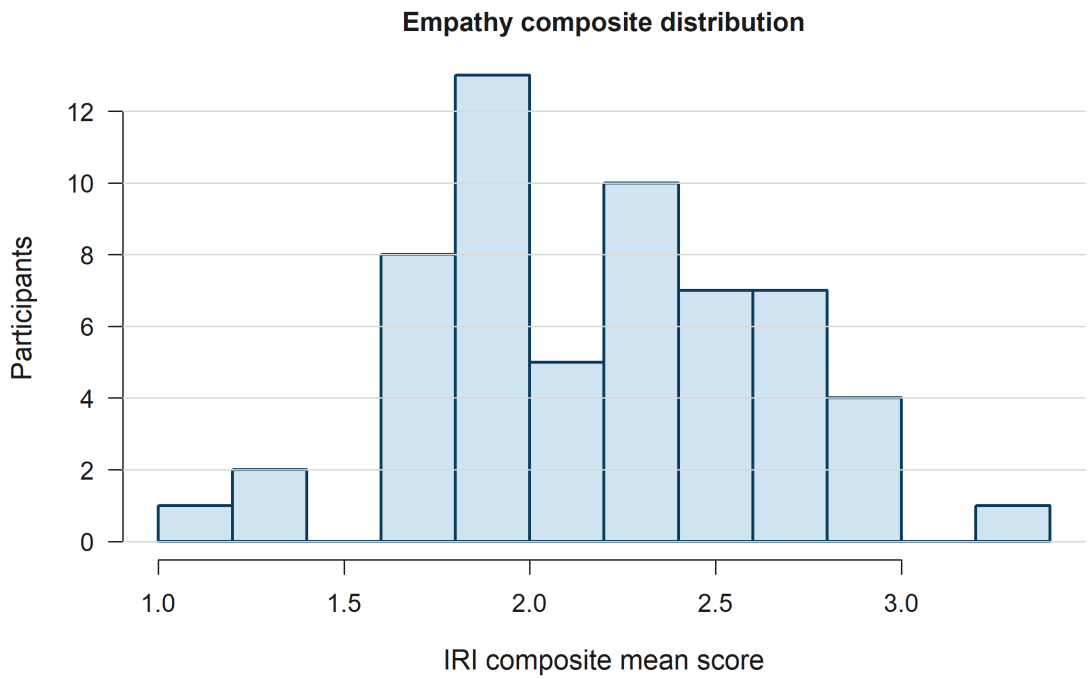

Figure 4: Empathy composite distribution.

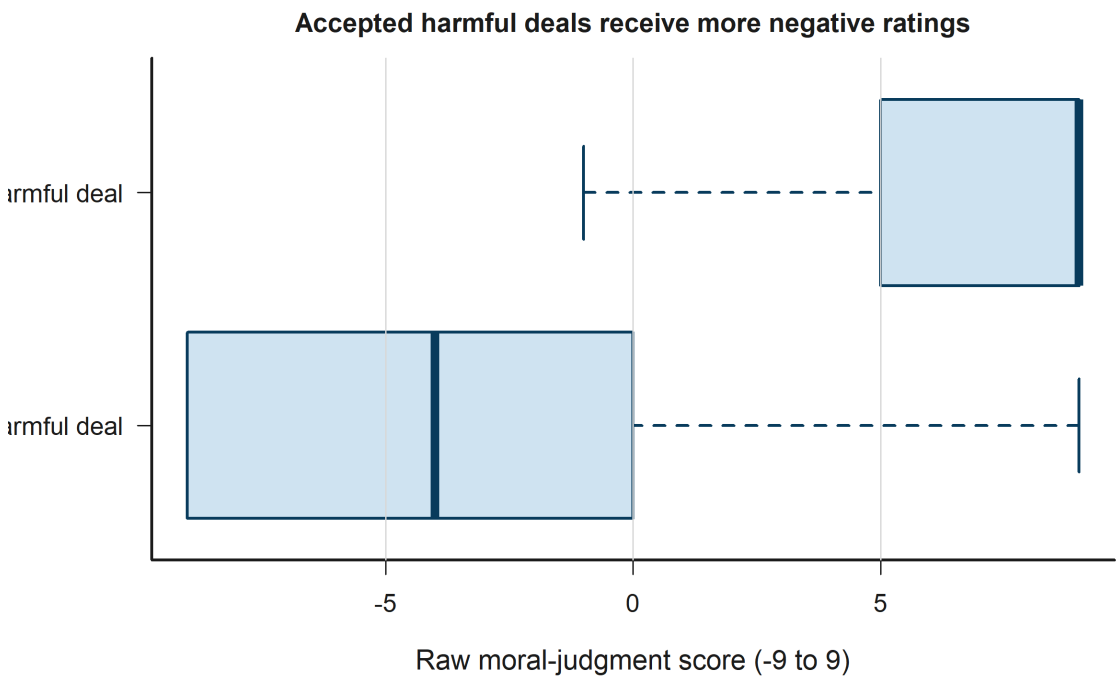

Figure 5: Raw moral judgment by decision type.

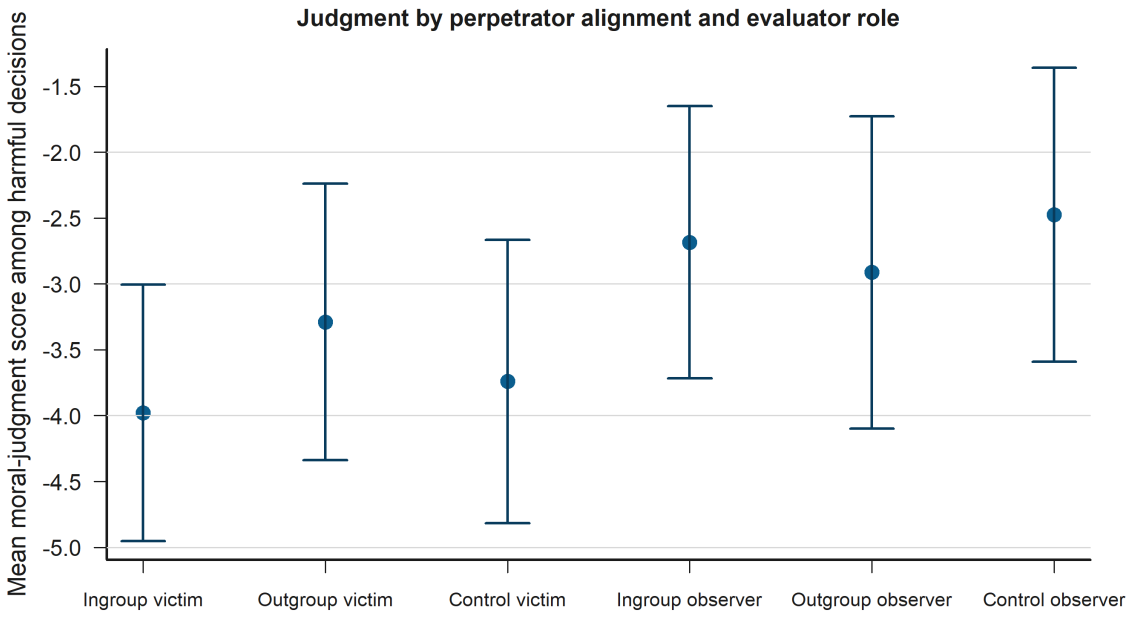

Figure 6: Mean judgment by perpetrator alignment and evaluator role.

3.3 Model Results

In the main harmful-decision Tobit model, higher empathy is associated with harsher judgments (b = -0.79, 95% CI [-2.86, 1.29], p = 0.457; not statistically distinguishable from zero). Outgroup perpetrators receive more favorable judgments than ingroup perpetrators (b = 0.08, 95% CI [-1.27, 1.44], p = 0.906; not statistically distinguishable from zero). The empathy slope becomes more negative in outgroup cases than ingroup cases (b = -0.45, 95% CI [-2.06, 1.16], p = 0.583; not statistically distinguishable from zero). In the same-faculty harm model, same-faculty harm is associated with harsher condemnation (b = -1.34, 95% CI [-2.88, 0.19], p = 0.087; not statistically distinguishable from zero). The main harmful-decision Tobit model indicates that empathy has a negative estimated association with judgments, consistent with harsher condemnation at higher empathy, but the estimate is imprecise and not statistically distinguishable from zero (b = -0.79, p = 0.457). The estimated outgroup-perpetrator effect is close to zero and slightly positive (b = 0.08, p = 0.906). The empathy-by-outgroup interaction is negative, which matches the theorized direction for H3, but it is also statistically inconclusive (b = -0.45, p = 0.583). Among the controls, outgroup victims are judged more negatively at the 0.05 threshold and observer-role evaluations are less negative than victim-role evaluations. Across the formal hypothesis tests, all four null hypotheses are retained at alpha = 0.05, although H2a is the closest case to the theorized pattern.

Table 6: Main harmful-decision Tobit coefficients.

Term	Estimate	RobustSE	CI_Low	CI_High	PValue
Intercept	0.508	5.090	-9.468	10.484	0.921
Empathy composite (z)	-0.788	1.059	-2.864	1.288	0.457
Outgroup perpetrator (ref = ingroup)	0.082	0.691	-1.272	1.436	0.906

Term	Estimate	RobustSE	CI_Low	CI_High	PValue
Control label hidden (ref = ingroup)	0.139	0.807	-1.443	1.721	0.863
Victim outgroup (ref = ingroup)	-2.417	1.233	-4.834	-0.000	0.050
Observer role (ref = victim)	2.890	1.394	0.158	5.622	0.038
Engineering participant (ref = humanities)	-1.279	1.801	-4.808	2.251	0.478
Man (ref = woman)	-1.821	1.761	-5.273	1.631	0.301
Age	-0.089	0.218	-0.516	0.338	0.683
Socioeconomic status	-0.435	0.760	-1.923	1.054	0.567
Stage 2 (ref = stage 1)	0.769	1.056	-1.301	2.840	0.466
Stage 3 (ref = stage 1)	-1.132	1.450	-3.973	1.710	0.435
Stage 4 (ref = stage 1)	-2.214	1.528	-5.208	0.781	0.147
Stage 5 (ref = stage 1)	-1.788	1.250	-4.238	0.662	0.153
Stage 6 (ref = stage 1)	-0.749	1.348	-3.391	1.893	0.578
Stage 7 (ref = stage 1)	-1.454	1.237	-3.878	0.970	0.240
Stage 8 (ref = stage 1)	-3.178	1.407	-5.936	-0.421	0.024
Stage 9 (ref = stage 1)	-2.733	1.358	-5.394	-0.071	0.044
Stage 10 (ref = stage 1)	-1.122	1.359	-3.787	1.542	0.409
Negotiator 2 (ref = negotiator 1)	0.002	0.485	-0.948	0.953	0.996
Empathy x outgroup perpetrator	-0.450	0.821	-2.059	1.159	0.583
Empathy x control label hidden	-0.450	0.733	-1.887	0.987	0.539

Table 7: Same-group-harm Tobit coefficients.

Term	Estimate	RobustSE	CI_Low	CI_High	PValue
Intercept	4.054	5.531	-6.788	14.895	0.464
Empathy composite (z)	-0.873	0.953	-2.740	0.994	0.359
Negotiator and victim share faculty	-1.341	0.783	-2.876	0.194	0.087
Outgroup perpetrator (ref = ingroup)	-0.578	0.721	-1.991	0.835	0.423
Victim outgroup (ref = ingroup)	-1.847	1.534	-4.854	1.160	0.229
Observer role (ref = victim)	2.294	1.520	-0.685	5.272	0.131
Engineering participant (ref = humanities)	-1.755	1.764	-5.211	1.702	0.320
Man (ref = woman)	-1.175	1.774	-4.652	2.302	0.508
Age	-0.213	0.227	-0.659	0.233	0.349
Socioeconomic status	-0.303	0.735	-1.743	1.137	0.680
Stage 2 (ref = stage 1)	-0.630	1.527	-3.623	2.363	0.680
Stage 3 (ref = stage 1)	-0.843	1.536	-3.855	2.168	0.583
Stage 4 (ref = stage 1)	-3.761	1.511	-6.722	-0.800	0.013
Stage 5 (ref = stage 1)	-1.283	1.822	-4.853	2.288	0.481
Stage 6 (ref = stage 1)	-0.760	1.393	-3.490	1.970	0.585
Stage 7 (ref = stage 1)	-2.578	1.478	-5.475	0.319	0.081
Stage 8 (ref = stage 1)	-2.125	1.470	-5.005	0.756	0.148
Stage 9 (ref = stage 1)	-1.470	1.362	-4.139	1.199	0.280
Stage 10 (ref = stage 1)	-1.463	1.611	-4.622	1.695	0.364
Negotiator 2 (ref = negotiator 1)	-0.147	0.576	-1.277	0.982	0.798

Table 8: Model fit summary.

Model	Observations	Participants	LowerBoundCensored	UpperBoundCensored	Lo
Main harmful-decision Tobit	569	58	170	27	-1
Same-faculty harm Tobit	364	58	103	13	-9
Full-sample Tobit	1160	58	182	332	-2

3.4 Hypothesis Tests and Decisions

Decision rule: reject H0 when $p < 0.05$; otherwise fail to reject H0. Support for the research hypothesis requires both statistical detectability and the theorized sign. H1: Fail to reject H0; Insufficient evidence for HA. H1 is inconclusive: fail to reject the null. The coefficient direction is negative but the estimate is not statistically distinguishable from zero. H2a: Fail to reject H0; Insufficient evidence for HA. H2a is inconclusive: fail to reject the null. The coefficient direction is negative but the estimate is not statistically distinguishable from zero. H2b: Fail to reject H0; Insufficient evidence for HA. H2b is inconclusive: fail to reject the null. The coefficient direction is positive but the estimate is not statistically distinguishable from zero. H3: Fail to reject H0; Insufficient evidence for HA. H3 is inconclusive: fail to reject the null. The coefficient direction is negative but the estimate is not statistically distinguishable from zero.

Hypothesis	Description	ModelTe
H1	Higher empathy predicts lower moral-judgment scores for harmful decisions.	Empathy
H2a	Same-faculty harm receives lower moral-judgment scores than cross-faculty harm.	Negotiat
H2b	Outgroup perpetrators receive lower moral-judgment scores than ingroup perpetrators.	Outgroup
H3	Empathy becomes more negative in outgroup than ingroup cases.	Empathy

4 Assumptions

The Tobit model is appropriate here because the observed dependent variable is bounded at -9 and 9 and the sample includes observations piled up at both limits. In the harmful-decision sample there are 569 negotiator-level observations from 58 participants. The observed censoring shares are 29.9% at the lower bound and 4.7% at the upper bound. Substantive interpretation assumes a latent continuous evaluation process, approximately normal model errors, and correct specification of the linear predictor. Inference is clustered by participant to address repeated judgments within persons, but the estimates should still be read cautiously because the participant count is modest.

5 Discussion

The results suggest that the study has some directional consistency with the empathy and ingroup-betrayal accounts, but the evidence is not strong enough to support the research hypotheses under a conventional 5 percent threshold. The clearest substantive pattern is the negative point estimate for same-group harm, which is compatible with ingroup betrayal, yet still too uncertain for a firm conclusion ($p = 0.087$). The lack of support for H2b indicates little evidence that outgroup perpetrators are condemned more harshly in this sample. Given the modest number of participants,

repeated judgments nested within persons, and substantial censoring at the lower bound, the null findings should be interpreted as limited evidence rather than strong evidence of no effect.

6 Conclusions

In conclusion, the bounded moral-judgment data are well suited to a Tobit framework, and the reporting pipeline now documents the full workflow from raw workbook to publication-style outputs. On the current data, the estimated effects do not justify rejecting the null hypotheses for H1, H2a, H2b, or H3 at alpha = 0.05. The most promising signal is the negative same-group-harm coefficient, which may warrant follow-up with a larger sample or a design with stronger identity contrasts.

7 Reproducibility Notes

The pipeline exports synchronized outputs in CSV, PNG, Markdown, Word, LaTeX, and PDF formats. The LaTeX and PDF outputs are generated directly from the fitted model results in the same run as the other reports.